



Figure 1: Emacs GUI mode

Figure 2: After running `emacs -nw`

However, our main interest in this article is the CLI mode, which you can start by running `emacs -nw` in the terminal. You should see something like what's shown in Figure 2.

Start a new file in Emacs by typing `emacs -nw filename` in the terminal. What you should see next is a blank buffer. Emacs works on the buffer system; anything you type is stored in a (volatile) buffer. When you save the buffer, it is written to permanent storage (probably your hard disk). The default buffer is known as the scratch buffer.

At the top is the menu bar, like in the GUI interface—but you hardly use it in the CLI. Below that is your buffer area, where you can input content or data. Below the buffer is the mode line. You can call this the status bar of the editor. It gives you important information about your Emacs session, and about the current buffer displayed above it, including: whether your latest changes were saved to disk, what line number the cursor is at, how far into the buffer the bottom of the screen is (in a percentage of the whole), and what Emacs features and settings are currently active. Below the mode line is the Mini Buffer, where Emacs displays messages relevant to ongoing operations.

Key combination	Function
C-x C-s	Saves the contents in the buffer to a file
C-x C-f	Finds a file, or starts a file by giving the location
C-x k	Kills the buffer
C-x b	Switches between multiple buffers
C-p	Moves the cursor to the previous line
C-n	Moves the cursor to the next line
C-f	Moves the cursor one letter forward
C-b	Moves the cursor one letter backward
M-f	Moves the cursor one word forward
M-b	Moves the cursor one word backward
C-a	Moves to the beginning of the current line
C-e	Moves to end of the line
M-a	Moves to the beginning of the sentence
M-e	Moves to the end of the sentence
C-_	Undoes the change (this is Ctrl+underscore, or Ctrl+Shift+minus)
C-space	Sets the mark for selecting the region
C-w	Used to kill the selected region (delete/cut)
C-y	Used to yank the selected region (undo/paste)
M-u	Converts the word under the cursor to upper-case
Alt-l	Converts the word to lower-case
C-x C-l	Converts a region to lower-case
C-z C-u	Converts a region to upper-case

Table 1: Useful Emacs key bindings

Key bindings

To free you from the need to use the mouse in CLI mode, there are many pre-defined key bindings (combinations and series of keystrokes that invoke a specific function) that let you invoke the functionality that you would have used the mouse or menus for, in the GUI mode. You can customise these bindings to your preference, but that is outside the scope of this article. In some cases, using a different Emacs *mode* can change the function that is invoked by a key sequence. We deal with *modes* below.

Many key bindings begin with two special keys: *Ctrl* and *Meta*. (The Meta key is usually the *Alt* key, or the *Esc* key.) For example, to exit Emacs and return to the command-line, the key sequence used is *Ctrl-x Ctrl-c* (holding the *Ctrl* key down, press *x* then *c*). In listings of key bindings, such as that at http://www.gnu.org/software/emacs/manual/html_node/emacs/Key-Index.html#Key-Index, these keys are abbreviated to 'C' and 'M' respectively; the same exit-Emacs sequence is listed at the above page as 'C-x C-c: Exiting'.

The mini buffer shows you the progress of your command as you are typing it.